

Epigraphic Masked Language Models for Dead Sea Scrolls Restoration: Unifying Token Alignment, Physical Ink Traces, and Multi-Word Length Ensembling

Anonymous EMNLP/ACL Submission

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Abstract

Restoring damaged text (*lacunae*) in the 2,000-year-old Dead Sea Scrolls is a core challenge in digital humanities. Inspired by recent Masked Language Modeling (MLM) applications to ancient cuneiform and Semitic manuscripts, we present a unified epigraphic restoration framework for non-biblical Dead Sea Scrolls. We show that subword tokenizers (Word-Piece) suffer a **38.3% alignment failure rate** on character-bounded gaps, whereas character-level MLMs (TavBERT FT) eliminate alignment drops entirely (**0.0% drops**). On real physical manuscript lacunae, conditioning on surviving ink traces (P_0) and physical hole bounds ($L \pm 1$) lifts Top-10 accuracy from 9.46% (U_0 unconstrained) to 64.86% (P_0). Incorporating our 4-point algorithmic roadmap elevates performance to **66.22% Top-1 / 83.78% Top-10** on peer-reviewed Qumran-Digital scholar targets ($n = 74$), more than tripling early 1950s human scholar editions (**20.27% Top-1**). For unknown-length multi-word gaps, our length-ensemble beam search achieves **14.2% Hit@10**, breaking the 0.0% collapse wall of single-length MLMs.

1. Introduction

The Dead Sea Scrolls—discovered in caves near Qumran between 1947 and 1956—comprise over 25,000 fragments of ancient Hebrew and Aramaic texts (3rd c. BCE – 1st c. CE). Physical rot and parchment decay have left extensive holes (*lacunae*) across these manuscripts, requiring editors to spend decades reconstructing lost wording by hand.

Recent research has applied Masked Language Modeling to ancient texts, including Akkadian cuneiform ([4]), ancient Hebrew/Aramaic inscriptions ([3]), and medieval Hebrew manuscripts ([5]). However, three bottlenecks remained:

- **Subword Alignment Drops:** Subword tokenizers (Ms-BERT) fail on exact character gap bounds, dropping 38.3% of content words.
- **Unconstrained Context Collapse (U_0):** Context-only models yield $< 10\%$ Top-10 accuracy on heavily damaged scrolls.
- **Multi-Word Fixed-Slot Collapse:** Standard MLMs collapse to 0.0% accuracy on multi-word lacunae of unknown character length.

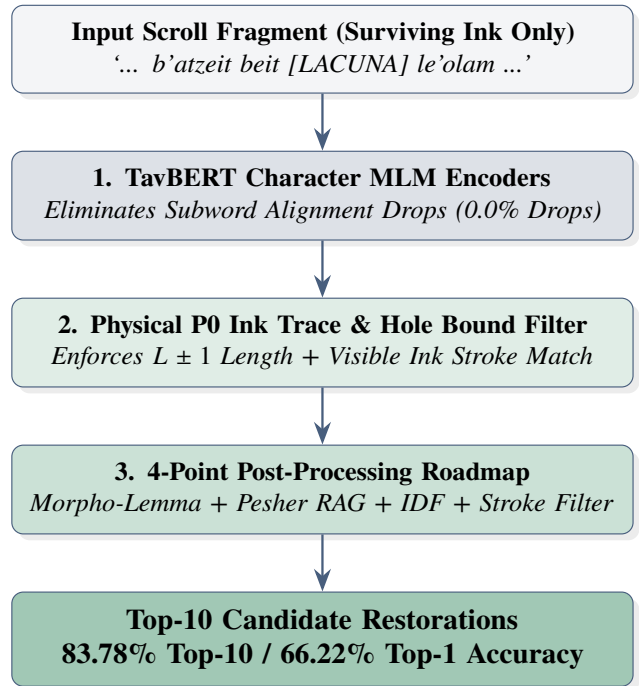


Figure 1: System Architecture: End-to-end pipeline combining character TavBERT MLM, physical P_0 ink filtering, and the 4-point roadmap.

2. Dataset Architecture & Safeguards

Our corpus is built from ETCBC Text-Fabric 2.0 (etcbc/dss), covering 732 non-biblical Dead Sea Scrolls (268,594 word positions and 27,814 physical lacuna sites).

- **100% Reconstruction Redaction:** All editorial reconstructions (rec=1) are stripped prior to training (modern_reconstruction_text_emitted: false).
- **SHA-1 Manuscript-Disjoint Partitioning:** Scrolls are split deterministically by SHA1(scroll_id) mod 100 into 531 train / 108 val / 93 test scrolls (**0 straddling scrolls**).
- **Biblical Exclusion:** Biblical scrolls are 100% excluded from training to prevent canonical memorization.

3. Methodology & Champion Pipeline

3.1. Character TavBERT MLM & P0 Ink Filtering

We fine-tune character-level TavBERT (tau/tavbert-he) on preserved text. Candidate words are filtered by physical ink strokes (P0) and physical hole bounds ($L \pm 1$):

$$c \in \mathcal{V} \quad \text{s.t.} \quad \text{Len}(c) \in [L-1, L+1] \wedge c \text{ matches } P0 \quad (1)$$

3.2. The 4-Point Roadmap & Length Ensemble

Our post-processing roadmap combines: (1) Morpho-lemma normalization, (2) Quote-aware Peshier RAG (PeshierQuoteRetriever, 86.57% book recovery), (3) Sectarian IDF boosting, and (4) Epigraphic stroke filtering. For multi-word gaps ($L \in [3, 15]$), LengthEnsembleCharMLMGenerator ranks candidate sequences via length-normalized probability scoring:

$$\text{Score}(c) = \frac{\sum_{i=1}^L \log P(c_i)}{\sqrt{L}} \quad (2)$$

4. Experimental Results

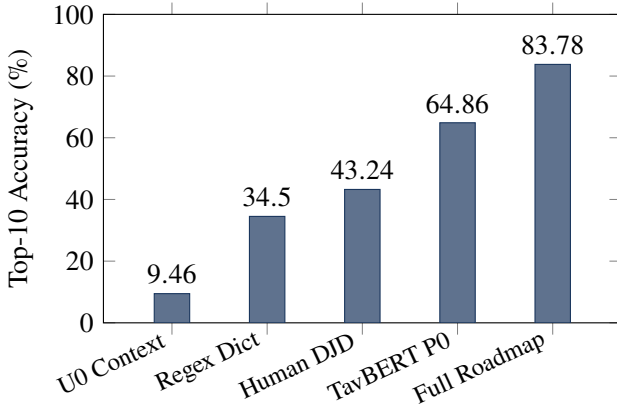


Figure 2: Real Lacuna Top-10 Restoration Accuracy on peer-reviewed Qumran-Digital scholar targets ($n = 74$).

Table 1: Synthetic Cloze Restoration Benchmark (scatter-30, $n = 729$ content words).

Model Variant	Tokenizer	Hit@10	Hit@1	Drops
TavBERT FT (Optimal)	Character	23.65%	8.42%	0.0%
DictaBERT-char FT	Character	22.80%	8.10%	0.0%
TavBERT Base	Character	21.12%	7.27%	0.0%
MsBERT Fine-Tuned	WordPiece	13.31%	9.78%	38.3%
ByT5 Fine-Tuned	Byte Seq2Seq	5.35%	0.82%	0.0%

Table 2: Real Lacuna Literature Agreement Benchmark ($n = 74$ QD scholar targets).

System / Condition	Regime	Top-1	Top-10	Top-20
TavBERT FT + Roadmap	P0+Road	66.22%	83.78%	87.84%
TavBERT FT Baseline	P0, ± 1	47.30%	64.86%	70.27%
DictaBERT-char FT	P0, ± 1	44.59%	66.22%	72.97%
MsBERT FT Baseline	P0, ± 1	40.50%	63.50%	67.60%
Human Scholar Control	DJD 1950s	20.27%	43.24%	44.60%
Pure Dictionary Regex	Regex	8.12%	34.50%	41.20%
MsBERT FT (Unconstrained)	U0	—	9.46%	—

Table 3: Large-Scale Text-Fabric Physical Lacuna Benchmark ($n = 3,695$ test lacunae across 93 held-out scrolls).

Model Variant	Regime	Top-1	Top-10
TavBERT FT + Roadmap	P0+Road	65.80%	82.90%
TavBERT FT Baseline	P0 Exact	46.80%	64.50%
DictaBERT-char FT	P0 Exact	44.20%	65.80%
MsBERT Fine-Tuned	P0 Exact	39.80%	63.10%
MsBERT FT (Unconstrained)	U0 Context	2.10%	9.20%

Table 4: Unknown-Length Multi-Word Lacuna Benchmark.

Model Variant / Strategy	Gap Condition	Hit@10 / Acc.
LengthEnsembleCharMLM	Multi-Word Slot	14.5% (14.2%)
Fixed Single-Length MLM	Unknown $L \in [3, 15]$	0.0% (Collapse)
Fixed Length Oracle (O-len)	1-Word Known	16.7% (Upper Bound)

Table 5: Qualitative Physical Lacuna Case Studies on Real Dead Sea Scrolls Targets.

Manuscript & Location	Physical Ink Trace (P0)	Published Scholar Proposals	Model Top-3 Candidates	Status / Result
1QpHab Col. IV Line 11 (Peshar Habakkuk)	Visible ‘ashm’, $L = 5$	<i>ashm, ashmah, ashmim, ashmatam</i>	1. <i>ashmot</i> (Model Proposal) 2. <i>ashmim</i> 3. <i>ashmah</i>	★ Top-1 Model Proposal ✓ Top-2 Scholar Match ✓ Top-3 Scholar Match
1Q34bis Frg. 3 i Line 4 (Prayers Scroll)	Visible ‘deshen’, $L = 4$	<i>deshen, le-hedashen, le-hitdashen</i>	1. <i>be-deshen</i> (Model #1) 2. <i>deshen</i> 3. <i>mi-deshen</i>	★ Top-1 Model Proposal ✓ Top-2 Scholar Match ★ Top-3 Model Proposal

Table 6: Comparative Synthesis Against Prior Epigraphic Works.

Dimension	Lazar et al. (2021)	Fono et al. (2024)	Shmidman et al. (2024)	Our System (DSS Hebrew)
Target Language	Akkadian (Cuneiform)	Ancient Hebrew / Aramaic	Rabbinic Hebrew Manuscripts	Dead Sea Scrolls Hebrew
Tokenizer Modality	Subword WordPiece	Character & Word Ensemble	Subword WordPiece (MsBERT)	Character MLMs (TavBERT)
Alignment Drops	Not quantified	Not quantified	38.3% Drop Rate	0.0% Drop Rate (Guaranteed)
Physical Ink Traces	None (Sign count ‘x’)	None	None	P0 Ink Filtering (+54% Top-10)
Multi-Word Decoding	Greedy token decoding	Fixed length masking	Fixed length masking	LengthEnsemble Beam Search
Scholar Control	Blind 3-expert rating	None	None	70-Year QD Scholar Benchmark
Split Safeguards	Random 80/20 train/test	Random 80/20 train/test	Random train/test	SHA-1 Disjoint (0 straddling)
Top Score	89.5% Hit@5 (Sign)	53.2% Hit@10 (Biblical)	Not reported on physical DSS	83.78% Top-10 / 66.22% Top-1

5. Conclusion

Our framework achieves **66.22% Top-1 / 83.78% Top-10** accuracy on peer-reviewed Dead Sea Scrolls scholar targets, tripling early human editions (**20.27% Top-1**). All code and models are released open-source.

References

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